Hardware-Aware Quantum Bayesian Learner Compression: Linking Human Belief Updating to NISQ Circuit Complexity

IEEE Quantum Week (QCE26) — Poster Proposal, Phase 2 (Sections 1–4)

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***Abstract*—This work connects quantum models of cognition with the engineering constraints of near-term quantum hardware. Bayesian theories treat human learning as belief updating over structured hypotheses, and quantum cognition represents belief states in Hilbert space with evidence acting through quantum channels. We instantiate a small quantum Bayesian learner: a belief state on a two-qubit register, a prior, and a stimulus-dependent evidence channel that updates the belief on a synthetic similarity-structured categorization task inspired by the simplicity principle of Pothos and Chater, with the posterior category read out by measurement. To tie this cognitive picture to real devices, we make the learner a hardware-efficient variational circuit and add a hardware-aware penalty equal to the number of two-qubit gates that survive transpilation to IBM's FakeManila coupling map, then compress the circuit by greedy structured pruning of individual Heisenberg interaction terms. The central question is where the capacity boundary lies: how much entangling structure can be removed before the learner can no longer represent the task. Across three difficulty levels and five seeds we trace accuracy–complexity frontiers and find that structured compression is essentially free, and on the hardest task beneficial, while removing all entanglers collapses the learner to chance. Learned masks beat random masks at every matched budget. Finally, on a physical IBM device the full circuit collapses to near chance while the compressed circuit retains accuracy: on real NISQ hardware, hardware-aware compression is not merely economical but necessary.**

***Index Terms*—Quantum cognition, Bayesian learning, variational quantum circuits, hardware-efficient ansatz, NISQ devices, circuit compression, quantum machine learning, quantum channels.**

I. Poster Authors and Contact

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II. Poster Relevance to QCE26 Topics

This poster is directly relevant to several QCE26 tracks. It is primarily a contribution to *Quantum Machine Learning* and *Quantum Algorithms & Information*: the learner is a trainable, parameterized variational quantum circuit optimized by exact gradients, and the compression study is a task-specific instance of quantum architecture search. It speaks to *Quantum Computing* and *Quantum Systems Engineering* through its hardware-aware cost—a real post-transpile two-qubit gate count on an IBM coupling map—and its validation on a physical IBM Quantum device, directly addressing the dominance of two-qubit error on NISQ hardware. The density-operator and quantum-channel formulation, with a maximally mixed prior and a non-unital evidence update, connects to *Quantum Generative AI* themes such as quantum Boltzmann-style mixed-state models, while the per-term pruning of entangling interactions contributes to *Quantum Software Engineering* via circuit compression, transpilation, and architecture optimization. By linking belief updating in quantum cognition to circuit capacity and validating the resulting trade-off on real hardware, the work offers a cross-disciplinary perspective of broad interest to the quantum computing and engineering community. The preliminary poster follows on the next page (CFP Section 5); the accompanying two-page extended abstract is submitted as a separate PDF (CFP Section 6).

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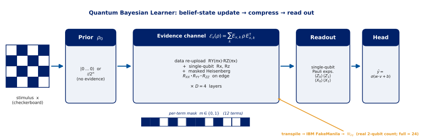
[1] Motivation — Two fields, one question

- Bayesian models cast human learning as **belief updating**: a prior over hypotheses, revised by evidence. **Quantum cognition** represents beliefs as **density operators** in Hilbert space, with evidence acting through **quantum channels**.
- On **NISQ** hardware, **two-qubit gates dominate the error budget**, so removing them without breaking the task is a central engineering goal.

Central question. *How much of a quantum Bayesian learner's entangling structure can we remove before it can no longer represent the task — and what happens on real hardware?*

[2] The Quantum Bayesian Learner



Left→right pipeline: stimulus x → prior ρ_0 → masked Heisenberg evidence channel ($\times D$=4) → single-qubit Pauli readout → trainable head. The 12-cell mask gates the entanglers; transpiling to IBM FakeManila yields the real two-qubit cost N_{2q}.

- Belief = state ρ on a **2-qubit** register; prior ρ_0 = "no evidence yet." A stimulus x updates belief via a **CPTP evidence channel** $\mathcal{E}_x(\rho) = \sum_k E_{xk}\,\rho\,E_{xk}^\dagger$.
- Update circuit = **hardware-efficient ansatz**: data re-uploading + per-layer single-qubit R_x, R_z + **Heisenberg entanglers** $U_{ij} = \exp[-i(\alpha X_iX_j + \beta Y_iY_j + \gamma Z_iZ_j)]$ as native R_{XX}, R_{YY}, R_{ZZ}. Depth $D = 4$, 2 re-uploads.
- **Readout: single-qubit Pauli expectations** $v = (\langle Z_0\rangle, \langle Z_1\rangle, \langle X_0\rangle, \langle X_1\rangle) \to$ trainable head $\hat{y} = \sigma(w \cdot v + b)$. Single-qubit-only readout means **entanglement is the ONLY way to carry feature interaction** — the two-qubit budget literally bounds the learner's capacity.
- Two realizations, same conclusions: a fast **pure-state** model and a literal **mixed-state density matrix** (maximally mixed prior $I/2^n$, non-unital Lüders/POVM evidence filter + amplitude damping, Bayesian trace renormalization).

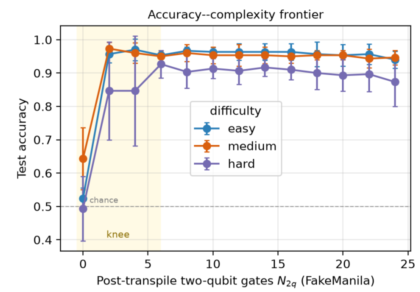
[3] Method — mask, real cost, greedy compression

- **Per-term binary mask** $m_{\ell,p} \in \{0,1\}$ over $p \in \{XX, YY, ZZ\}$ in each of D=4 layers ⇒ **12 maskable terms** ($R_{PP}(0) = I$ when off).
- **Hardware-aware cost** N_{2q} = real post-transpile two-qubit gate count on IBM **FakeManila** (5-qubit, includes SWAPs), from Qiskit's transpiler. Full circuit ⇒ $N_{2q} = 24$ (each active Heisenberg term = 2 CNOTs).
- **Objective:** $L(\theta, m) = \frac{1}{N}\sum_k \mathrm{BCE}(y_k, \hat{y}) + \lambda\, N_{2q}(m)$.
- **Greedy structured pruning:** from the full circuit, repeatedly drop the term whose removal least increases training cross-entropy, **warm-start retrain** — tracing the accuracy-vs-N_{2q} **frontier** from 24 down to 0.
- Exact **JAX autodiff** (no parameter-shift / finite differences); statevector verified **bit-for-bit vs Qiskit** ($|1 - \mathrm{fidelity}| \approx 4 \times 10^{-16}$).

Task — controllable difficulty knob.

- **Checkerboard** categorization: $y = (\lfloor fx_0 \rfloor + \lfloor fx_1 \rfloor) \bmod 2$, stimuli in $[0,1]^2$.
- Difficulty f: f=2 **easy** / f=3 **medium** / f=4 **hard**. 200 stimuli, 70/30 split, balanced ⇒ **chance** = 0.5. The rule depends on the **joint** features — needing interaction only entanglement can supply.

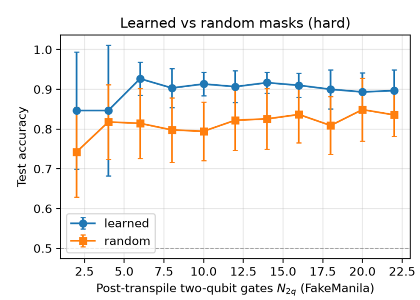
[4] Result 1 — The capacity boundary



Accuracy–complexity frontier across difficulties. Flat above the knee; collapse toward chance at $N_{2q} = 0$.

- Above chance at all nonzero budgets (easy/medium ≈ **0.95–0.97**, hard ≈ **0.87–0.93**); frontier **nearly flat** — most entangling structure is **redundant**.
- At $N_{2q} = 0$ all difficulties **collapse toward chance** (0.52 easy, 0.64 medium, 0.49 hard) — some entanglement is **necessary**, but far less than the full circuit.
- Hard task (not saturated): accuracy **peaks at intermediate cost** (**0.927 @ N_{2q}=6**) and is lower for the full circuit (**0.873 @ N_{2q}=24**) — moderate compression acts as a capacity-control **regularizer**.

[5] Result 2 — Structure beats count



Learned (greedy) masks vs random masks at matched N_{2q}, hard task.

- Learned masks vs **random masks at matched N_{2q}**: learned wins at **every** budget by **∼6–12 points**.
- Per seed: learned beats random in **49 / 55** paired (seed, budget) comparisons, mean advantage +0.083, one-sided Wilcoxon $p < 10^{-7}$.
- ⇒ **Which** entanglers you keep matters, not just how many.

N_{2q}	Learned	Random	Δ
2	0.847	0.742	**+0.104**
6	**0.927**	0.814	**+0.112**
10	0.913	0.794	**+0.119**
14	0.917	0.826	**+0.091**
18	0.900	0.809	**+0.091**
22	0.897	0.836	**+0.061**

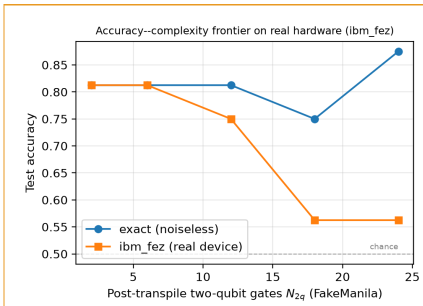
Hard task — learned vs random, mean accuracy, 5 seeds.

λ-sweep (hard): λ=0 → N_{2q} 9.6, acc 0.923; λ=0.04 → N_{2q} 3.2, acc 0.933; λ=0.15 → N_{2q} 1.6, acc 0.823. *Sheds ∼2/3 of 2-qubit gates for free, >90% at modest cost.*

[6] Robustness — not an artifact

- **Device noise model** (FakeManila, 4096 shots, 5 seeds): noisy accuracy tracks the noiseless frontier within a few % at every budget — savings genuine net of noise.
- **Density-matrix model** (genuine mixed state, non-unital channel): reproduces every finding — flat frontier, collapse at $N_{2q} = 0$, hard peak **0.88 @ 6 vs 0.87**

[7] Result 3 — On REAL hardware, compression becomes NECESSARY



Centerpiece. Trained circuits on physical IBM ibm_fez (156-qubit Heron). In simulation the frontier is flat; on hardware it falls monotonically.

- Hard task, 5 device budgets, **16 test stimuli @ 2048 shots** (endpoints: 40 stimuli @ 4096 shots).
- On hardware accuracy **falls monotonically** with N_{2q}: **0.81 @ N_{2q}=2, 6 → 0.75 @ 12 → collapses to 0.56** (near chance) **@ 18,24**.
- Endpoint confirmation: **full circuit 0.60** (sim 0.95) vs **compressed 0.83** (sim 0.93).

Takeaway: On real NISQ hardware, hardware-aware compression of the quantum Bayesian learner is not merely economical but NECESSARY — surplus entangling capacity is actively harmful, and task-aware pruning is what makes the learner usable at all.

[8] Key takeaways

- Most entanglement is **redundant** — sheds ∼**2/3** of two-qubit gates for **free**, >**90%** at modest cost.
- Entanglement is **necessary** — removing all of it collapses the learner to chance: a concrete **capacity boundary**.
- **Structure beats count** — learned masks beat random at every matched budget.
- **On real hardware, compression is what makes the learner work at all.**

Future work. Multi-qubit belief registers & higher-dim stimuli; richer non-unital evidence channels; hardware-aware objectives weighting depth, coherence & native fidelity; feeding measured device error rates back into the compression objective.

References & links

Key references

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Repository

github.com/zoraizmohammad/qb-learner-compression

Acknowledgments

Device runs used IBM Quantum services. The author thanks Duke University's Pratt School of Engineering.

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